

YAWER NAZIR RATHER

Computer Science & Engineering | Cybersecurity · Artificial Intelligence
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ABOUT ME

Computer Science and Engineering graduate interested in cybersecurity, artificial intelligence and quantum cryptography. Experience spans machine learning research, secure systems, software development and embedded applications, with a focus on applying computational methods to practical problems and building reliable, intelligent technologies.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

2022 - 2026

Central University of Kashmir, Srinagar, India

- **Percentage: 74%.**
- **Relevant coursework:** Computer Programming, Object Oriented Programming, Data Structures, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Computer Networks, Network Security, Artificial Intelligence, Machine Learning and Deep Learning.
- **Programming languages:** C, C++, Java and Python.
- **Mathematics:** Mathematics I and II, Mathematics III (Differential Equation), Mathematics IV (Probability and Statistics), and Discrete Structures.

Higher Secondary School Examination (Class XII - Science/PCM)

Completed 2020

Space Age Higher Secondary School, Srinagar, India

- Percentage: 82% | Grade: 8.6

Secondary School Examination (Class X)

Completed 2018

New Generation Public School, Srinagar, India

- CGPA: 9.8 / 10.0

RESEARCH EXPERIENCE

Research Intern - Deep Learning & Biometrics

Dec 2025 - Present

Indian Institute of Technology Jammu, India | Supervisor: Dr. Karan Nathwani

- Developed deep learning models for multimodal classification using the MM-AAD dataset, with an emphasis on representation learning and adaptation across recording conditions.
- Built a Cross-Stimulus Autoencoder (CSAE) to align audiovisual EEG representations with the feature distribution expected by a frozen, pretrained DARNet classifier, enabling adaptation without retraining the classifier.
- Trained and evaluated models in PyTorch across 30 labelled participants; compared random-window and trial-tail partitions, with identity initialization and control experiments to assess the learned representation mapping.
- Worked with iris and voice datasets for multimodal biometrics.

PUBLICATIONS & RESEARCH MANUSCRIPTS

Accepted Conference Paper

- Yash Paul, Shahid Sultan, Nazima Yousuf, **Yawer Nazir Rather**, and Wikas Khursheed. "An Efficient Graph-Mamba Network for EEG Auditory Attention Decoding." **Accepted**, IEEE IC2E3 2026, NIT Uttarakhand.

Co-authored a deep learning architecture combining hypergraph neural networks, Mamba state-space models and contrastive representation learning for classification across subjects.

Submitted Manuscript

- Rantu Buragohain, **Yawer Nazir Rather**, Sunil Kumar Kopparapu, and Karan Nathwani. "Cross-Stimulus EEG Representation Alignment for Auditory Attention Decoding Across Sessions." **Submitted to ICASSP 2027.**

CERTIFICATIONS & HONORS

- **Certificate of Appreciation - Session Moderator, 2AI 2026:** International Conference on Applied Artificial Intelligence, Central University of Kashmir, June 2026.
- **Building RAG Agents with LLMs - NVIDIA,** December 2025.
- **Certified in Cybersecurity (CC) - ISC2.**
- **Cybersecurity Fundamentals - HarvardX (edX).**
- **Arduino Timers & Interrupts - Udemy.**

PROJECTS

Beyond Signatures: Semantic Intelligence Against Generative Malware Evolution

2026

B.Tech Final-Year Project | Cross-Modal Transformer with Multi-Head Latent Queries (CMT-MHLQ)

- Developed an AI-based malware detector using a Cross-Modal Transformer with Multi-Head Latent Queries to identify malicious behaviour despite source-code rewriting and LLM-generated variants.
- Fused system-call sequences, process-resource graphs, process-lineage graphs and statistical profiles with Transformer/graph encoders and latent-query cross-attention; collected behaviour using Xen, DRAKVUF, LibVMI and AltP2M.
- Implemented generation-balanced training, orthogonality regularisation and collector-invariance learning; evaluated across five seeds and five folds (25 checkpoints), with family-disjoint splits and parent/sibling leakage controls.
- Achieved **0.9979 ± 0.0006 ROC-AUC** and **97.25% true-positive rate at a 1% calibration false-positive rate** on standard detection; retained **96.98% ± 10.50% recall** on source-rewritten L4 variants and **98.81% ± 0.73%** on cross-language LLM-generated L5 variants.
- Conducted held-out-family zero-day evaluation, modality ablations, collector-bias analysis and behavioural-integrity checks to assess robustness and generalization.

Cloud-Native Malware Analysis Lab

Oct 2025

- Deployed a Docker-based malware sandbox on Google Cloud Platform; automated sample execution, network-traffic capture and indicator-of-compromise extraction, with ELK Stack visualizations.

Kernel Watchtower: eBPF-Based Rootkit Detection

May - Jun 2025

- Built an eBPF/BCC monitoring tool to trace suspicious system calls and process execution, with a Python userspace agent for alert logging and threat visualization.

Smart Parking Management System

Aug - 2024

- Built an Arduino Uno prototype in C++ using an HC-SR04 ultrasonic sensor to detect parking-space occupancy, an RGB status indicator and a 16×2 I2C LCD for live monitoring.
- Implemented three-reading validation to reduce false occupancy changes from transient sensor noise, with visual guidance for available and occupied spaces.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, Java, SQL, Bash, PHP and 8086 Assembly.
- **Machine learning:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, PyTorch Geometric, graph neural networks, MNE-Python and OpenCV.
- **NLP and LLMs:** Retrieval-augmented generation, BERT, semantic search and vector databases.
- **Security and systems:** Xen, DRAKVUF, LibVMI, eBPF/BCC, Wireshark, Ghidra, Metasploit, Linux, Docker, Kubernetes, Git, GCP and AWS.
- **Methods and embedded systems:** Representation learning, multimodal fusion, statistical evaluation, optimization, Arduino and embedded C++.

LEADERSHIP & ACTIVITIES

- **Institute Ambassador, DEFCON Srinagar:** Organized capture-the-flag competitions and security workshops for 100+ students; represented Central University of Kashmir.
- **Head, University Research Club:** Coordinated research workshops and seminars; mentored peers in research methods and academic writing.
- **Event Coordinator, Cyber Conclave 2025:** Managed logistics and speaker sessions for the university cybersecurity summit.
- **National Service Scheme:** Led Swachh Bharat Abhiyan activities and coordinated rural blood donation camps.

LANGUAGES

- **English:** C1 (listening, reading, writing and speaking). **Urdu and Kashmiri:** Native / bilingual proficiency.